

## Additional Mathematics

## Chapter 10: Trigonometric Functions

## NOTES

## 10.1 Trigonometric Functions of Angles

|        | $0^\circ$            | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$           |
|--------|----------------------|----------------------|----------------------|----------------------|----------------------|
| $\sin$ | $\frac{\sqrt{0}}{2}$ | $\frac{\sqrt{1}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{4}}{2}$ |
| $\cos$ | $\frac{\sqrt{4}}{2}$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{1}}{2}$ | $\frac{\sqrt{0}}{2}$ |
| $\tan$ | 0                    | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | —                    |

## ① Degrees to Radians conversion

$$1^\circ = \frac{\pi}{180} \text{ rad} \quad 1 \text{ rad} = \frac{180^\circ}{\pi}$$

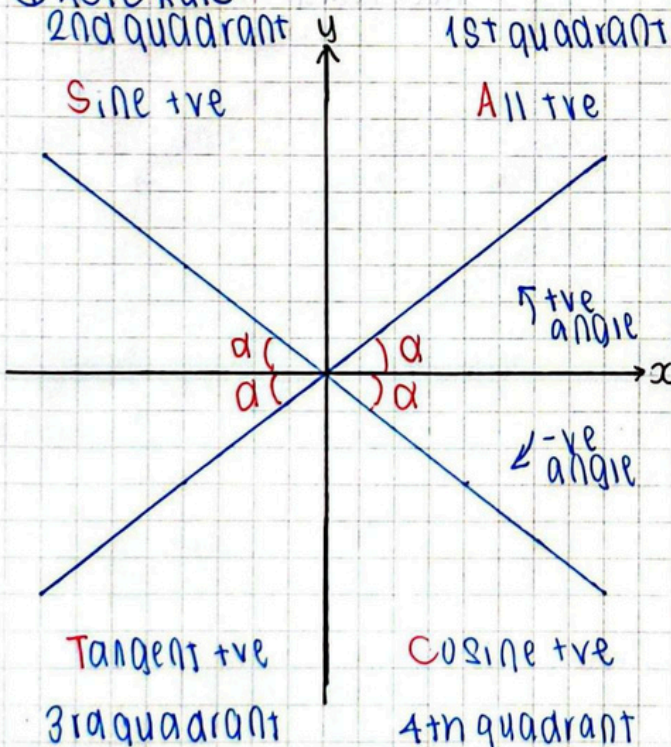
② Relationship of  $\theta$  and  $90^\circ - \theta$ 

- $\sin(90^\circ - \theta) = \cos \theta$
- $\cos(90^\circ - \theta) = \sin \theta$
- $\tan(90^\circ - \theta) = \frac{1}{\tan \theta}$

## ③ Reference Angles

- 1st quadrant  $\rightarrow \alpha = \theta$
- 2nd quadrant  $\rightarrow \alpha = 180^\circ - \theta$  or  $\pi - \theta$
- 3rd quadrant  $\rightarrow \alpha = \theta - 180^\circ$  or  $\theta - \pi$
- 4th quadrant  $\rightarrow \alpha = 360^\circ - \theta$  or  $2\pi - \theta$

## ④ ASTC Rule



## ⑤ Negative Angles

- $\sin(-\theta) = -\sin \theta$
- $\cos(-\theta) = \cos \theta$
- $\tan(-\theta) = -\tan \theta$

## ⑥ cosecant, secant, cotangent

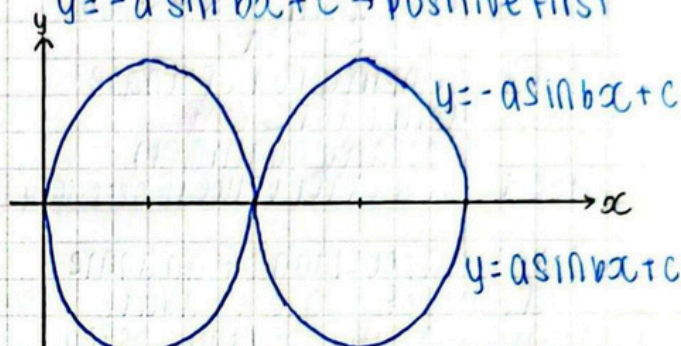
- $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\cot \theta = \frac{1}{\tan \theta}$

## 10.2 Graphs of Trigonometric Functions

## ⑦ Sine Graphs

$$y = a \sin bx + c \rightarrow \text{negative first}$$

$$y = -a \sin bx + c \rightarrow \text{positive first}$$

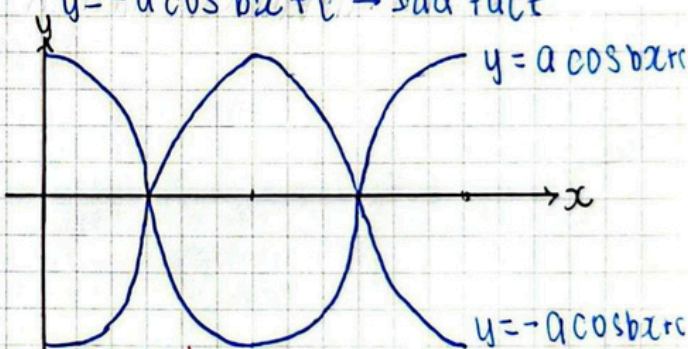


Amplitude =  $a$       Min value =  $-a + c$   
 Period =  $\frac{360^\circ}{b}$  or  $\frac{2\pi}{b}$       Max value =  $a + c$   
 Axis:  $y = c$

## ⑧ Cosine Graphs

$$y = a \cos bx + c \rightarrow \text{happy face}$$

$$y = -a \cos bx + c \rightarrow \text{sad face}$$

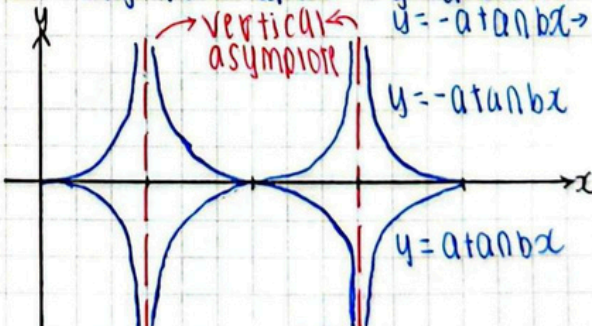


Amplitude =  $a$       Min value =  $-a + c$   
 Period =  $\frac{360^\circ}{b}$  or  $\frac{2\pi}{b}$       Max value =  $a + c$   
 Axis:  $y = c$

## ⑨ Tangent Graphs

$$y = a \tan bx \rightarrow \text{tve first}$$

$$y = -a \tan bx \rightarrow \text{ve first}$$



Period =  $\frac{\pi}{b}$  or  $\frac{180^\circ}{b}$   
 Vertical asymptote =  $\frac{\pi}{2b}$  or  $\frac{90^\circ}{b}$

## Principal values

Principal values of  $\sin^{-1} x$ :  $-90^\circ \leq x \leq 90^\circ$

Principal values of  $\cos^{-1} x$ :  $0^\circ \leq x \leq 180^\circ$

Principal values of  $\tan^{-1} x$ :  $-90^\circ < x < 90^\circ$

## QUESTION TYPES

### 10.1 Trigonometric Functions of Angles

#### 1. Simplifying trigonometric functions to reference angle $\alpha$

Step 1: Change negative angles to positive angles ⑥

Step 2: Minus  $360^\circ$  or  $2\pi$  from angle until  $0^\circ \leq \theta < 360^\circ$  OR  $0 \leq \theta < 2\pi$

Step 3: Simplify to ref angle using ASTC ③④ (Identify quadrant)

Step 4: Find exact value if necessary.

#### 2. Given $\sin/\cos/\tan$ of an angle and range of angle, evaluate values of $\sin/\cos/\tan/\operatorname{cosec}/\sec/\cot$

Step 1: Identify quadrant of angle using ASTC ④

Step 2: Draw ASTC graph with angle

Step 3: Label sides, use Pythagoras' Theorem to find last side

Step 4: Evaluate values of functions.

### 10.2 Graphs of Trigonometric Functions

#### 3. Find amplitude and axis given min and max value.

$$\text{Amplitude} = \frac{\text{Max value} - \text{Min value}}{2}$$

$$\text{Axis: } y = \frac{\text{Min value} + \text{Max value}}{2}$$

#### 4. Find number of solutions of complicated trigo expression by sketching graph

Step 1: Rearrange terms to get: expression 1 = expression 2

Step 2: Solutions are number of intercepts of  $y = \text{expression 1}$  and  $y = \text{expression 2}$

#### 5. Find principal values of trigonometric functions.

Step 1: Key in calculator (DEGREES MODE)

Step 2: Change to radians if necessary. ①

## Additional Mathematics

## Chapter 11: Trigonometric Identities and Equations

## NOTES

## 11.1 Simple Trigonometric Identities

## ① Tangent and cotangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

## ② Pythagorean Identities

$$1. \sin^2 \theta + \cos^2 \theta = 1$$

$$2. 1 + \tan^2 \theta = \sec^2 \theta$$

$$3. 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

## 11.2 Solving Trigonometric Equations

③  $\theta$  from  $\alpha$ 

$$1^{\text{st}} \text{ quadrant} \rightarrow \theta = \alpha$$

$$2^{\text{nd}} \text{ quadrant} \rightarrow \theta = 180^\circ - \alpha \text{ OR } \pi - \alpha$$

$$3^{\text{rd}} \text{ quadrant} \rightarrow \theta = 180^\circ + \alpha \text{ OR } \pi + \alpha$$

$$4^{\text{th}} \text{ quadrant} \rightarrow \theta = 360^\circ - \alpha \text{ OR } 2\pi - \alpha$$

## 11.3 Addition Formulae

## ④ Addition Formulae

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

## 11.4 Double Angle Formulae

## ⑤ Double Angle Formulae

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

## ⑥ Half Angle Formulae

$$\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$\cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2}$$

$$= 2 \cos^2 \frac{A}{2} - 1$$

$$= 1 - 2 \sin^2 \frac{A}{2}$$

$$\tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}$$

## 11.5 R-Formulae

## ⑦ R-Formulae

$$a \sin \theta \pm b \cos \theta = R \sin(\theta \pm \alpha)$$

$$a \cos \theta \pm b \sin \theta = R \cos(\theta \mp \alpha)$$

$$\text{where } \alpha \text{ is acute, } R = \sqrt{a^2 + b^2},$$

$$\tan \alpha = \frac{b}{a}$$

## QUESTION TYPES

## 11.2 Solving Trigonometric Equations

1. Solving Trigonometric Equations in the form  $\operatorname{trig}(ax+c) = n$ 

Step 1: If  $n = -1, 0$  or  $1$ , visualise the graph then set  $ax+c$  to the value when graph is at  $n$ .

Step 2: Find reference angle

$$\text{Reference angle, } \alpha = \operatorname{trig}^{-1}(|n|)$$

Step 3: Identify quadrants:

$ax+c$  is in the (quadrant) or (quadrant) since  $\operatorname{trig}(ax+c) = ?$

Step 4: Extend range.

(KEEP IN MIND DEG/RAD)

$$0 \leq x \leq k \rightarrow c \leq ax+c \leq ak+c$$

Step 5: With the new range, add/minus  $360^\circ$  or  $2\pi$  from each solution until out of range.

Step 6: Minus  $c$  on each solution then divide by  $a$  to solve for  $x$ .

2. Solving Quadratic Trigo Equations in the form  $\operatorname{trig}^2(ax+c) = n$ 

Step 1:  $\operatorname{trig}(ax+c) = \pm \sqrt{n}$

Step 2: solve like Q1. Except for step 3,  $ax+c$  can be in all 4 quadrants.

3. Solving Quadratic Trigo Equations in the form  $ax^2+bx+c=0$ , where  $x$  is a trig function.

Step 1: Factorise / Formula

Step 2: Reject if not  $-1 \leq x \leq 1$  for sine & cosine.

## 11.3 Addition Formulae

## 4. Find exact value of trig function using addition formulae.

Step 1: Identify how the expression can be rewritten as sum / difference of 2 easy values.  $30^\circ, 45^\circ, 60^\circ, 90^\circ, 180^\circ, 270^\circ$  if not, use double angle formula instead.

Step 2: Apply addition formulae. ④

#### 11.4 Double Angle Formulae

5. Find exact value of trig function using double angle formulae.

Step 1: Use double angle formulae to multiply value by 2.

consider:

$$\text{trig}(2\theta) = \text{trig } 2(\theta)$$

Step 2: simplify  $\text{trig}(2\theta)$  using ASC, then expand RHS using double angle formulae.

Step 3: Simplify. If square root is necessary, identify quadrant and reject other solution.

#### 11.5 R-Formulae

6. Simplifying trig functions using R-Formulae.

$$\text{Let } a \sin \theta \pm b \cos \theta = R \sin(\theta \pm \alpha),$$

where  $R > 0$  and  $\alpha$  is acute.

Step 1: Find  $R$  using  $R = \sqrt{a^2 + b^2}$

Step 2: Find  $\alpha$  using  $\tan^{-1}(\frac{b}{a})$

7. Find min/max values of  $R \sin(\theta \pm \alpha) + c$  or  $R \cos(\theta \pm \alpha) + c$  and corresponding values of  $\theta$ .

$$\text{Min value} = -R + c$$

when  $\sin/\cos = -1$ ,

$$\text{Max value} = R + c$$

when  $\sin/\cos = 1$

Step 1: Use graph to deduce angle.

If  $R < 0$ ,  $-\sin \theta / -\cos \theta$  graphs

If  $R > 0$ ,  $\sin \theta / \cos \theta$  graphs

Step 2: Extend range, list solutions of  $\theta \pm \alpha$  (from deduced angle)

Step 3: Find  $\theta$  by moving  $\alpha$  to RHS.

#### 11.6 Applications of Trigonometric Functions

8. Ferris wheel

Step 1: Axis of graph,  $c$

→ height of centre from ground

Step 2: Amplitude,  $a$  of wheel

→ radius of wheel

Step 3: Period,  $\frac{2\pi}{\omega}$

→ Time taken for one complete revolution

#### Proving Questions

Format:

$$\text{LHS/RHS} = \dots$$

$$\dots = \text{RHS/LHS}$$

Start from more complex side!

Step 1: Identify:

1. Convert  $\tan, \cot, \csc, \sec \rightarrow \sin/\cos$

2.  $\sin^2 x, \cos^2 x \rightarrow$  Pythagorean identity

3.  $\sin 2x, \cos 2x, \tan 2x \rightarrow$  Double Angle

4.  $1 \pm \sin x$  OR  $1 \pm \cos x$

→ Rationalise using conjugate,

then use Pythagorean identity.

Step 2: Think of how to make the terms on the other side.

Strategy 1: Split fraction

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$$

Strategy 2: Multiply top and bottom to be able to use

Pythagorean, then split

$$\frac{c}{x} = \frac{c^2}{cx} = \frac{1-s^2}{cx} = \frac{1}{cx} - \frac{s^2}{cx} \dots$$